

EDUCATION

Bachelor of Engineering Science, Mechatronics Engineering with AI Specialization

Western University | June 2026 | GPA: 3.70/4.00 | Dean's List 2022, 2023, 2024, 2025

- Capstone: Autonomous garbage collection and sorting robot using ROS2 for navigation and control, a robotic arm for object pickup and sorting, and computer vision (OpenCV, Roboflow) for real-time classification
- 4th Year Project: Simulated automated wheelchair control using ROS2 and Raspberry Pi for sensor integration, path planning, and real-time decision making

TECHNICAL SKILLS

AI/ML: OpenCV, TensorFlow, PyTorch, Scikit-learn, Roboflow, n8n, RAG, NLP, LLM fine-tuning

Programming: Python, SQL, Microsoft VBA, C, JavaScript, RESTful APIs, AWS Lambda

Engineering: ROS2, MATLAB, SolidWorks (CSWA Certified), control systems, thermal analysis

Tools & Platforms: AWS (Lambda, API Gateway), GCP, Raspberry Pi, Arduino, SAP, Jira

Certifications: Microsoft AI-900 (Azure AI Fundamentals) & SC-900 (Cybersecurity), SolidWorks CSWA

EXPERIENCE

Product Engineer

Scooty Mobility | May 2025 – Present

- Developing an AI-powered transit companion app for the Greater Toronto and Hamilton Area, featuring conversational trip planning and cross-agency routing across TTC, GO Transit, and MiWay
- Building AI-powered customer service automation using workflow orchestration tools to route support tickets and automate response handling
- Developing RESTful APIs in Python to simulate PRESTO card-based fare transactions for mobility applications
- Architecting serverless backend using AWS Lambda and API Gateway for scalable, event-driven processing
- Created AI in Finance training curriculum in partnership with Ontario Tech University, designed 5 weeks of learning materials, onboarded 80 participants, and prepared students for Microsoft SC-900 certification

Thermodynamics Engineering Intern

Bombardier Aerospace | Summers 2022 & 2023

- Core developer on Global Companion, a finalist-award-winning AI chatbot that dynamically sequences aircraft testing procedures based on part availability and provides instant troubleshooting guidance
- Built automated data pipelines using VBA to transform raw parameters into simulation-ready input files, eliminating manual data entry for thermal calculations
- Performed data analysis using SQL and Python to extract, clean, and validate temperature data from flight test datasets
- Developed heat transfer modeling tools to analyze thermal system behavior across aircraft surfaces

Engineering Summer Student

Thurber Engineering | Summers 2020 & 2021

- Conducted geotechnical laboratory testing and digitized technical documentation systems